

Assignment :- 2

Real-World Application: Online Video Streaming Platforms (e.g., Youtube, Netflix).

1/ Use of Parallel Computing: Parallel computing is used in video encoding, recommendation systems, & real-time analytics.

- Video Encoding: Videos are split into chunks & encoded in parallel using GPU & multi-core CPUs. This speeds up processing for large volumes of uploaded content.
- Recommendations: AI models for suggesting content use parallel processing to handle massive datasets quickly.
- Analytics: Real-time viewership data is processed in parallel to monitor performance & trends.

Why it's important:

- Reduces latency.
- Increases throughput & processing speed.
- Enables scalable, real-time processing.

2/ Use of Networked Systems: Networked systems deliver video content from distributed servers (CDNs - Content Delivery Networks) to users worldwide.

- Client-server architecture: Clients (users) request video; servers deliver based on location.
- Load balancing: Traffic is distributed among servers to prevent overload.
- Reliability: If one server fails, others handle the request seamlessly.

Why it's important:

- Ensures global availability & fast delivery.
- Enhances user experiences with minimal buffering.
- Supports millions of concurrent users efficiently.